
Department of Electrical & Computer Engineering
Cullen College of Engineering
University of Houston
ECE-3441: Digital Logic Design
Fall 2022 – Lab Exam

RULES:

1. You are not allowed to collaborate with others and browse online for solutions.
2. Open book, open notes.

DESIGN PROBLEM:

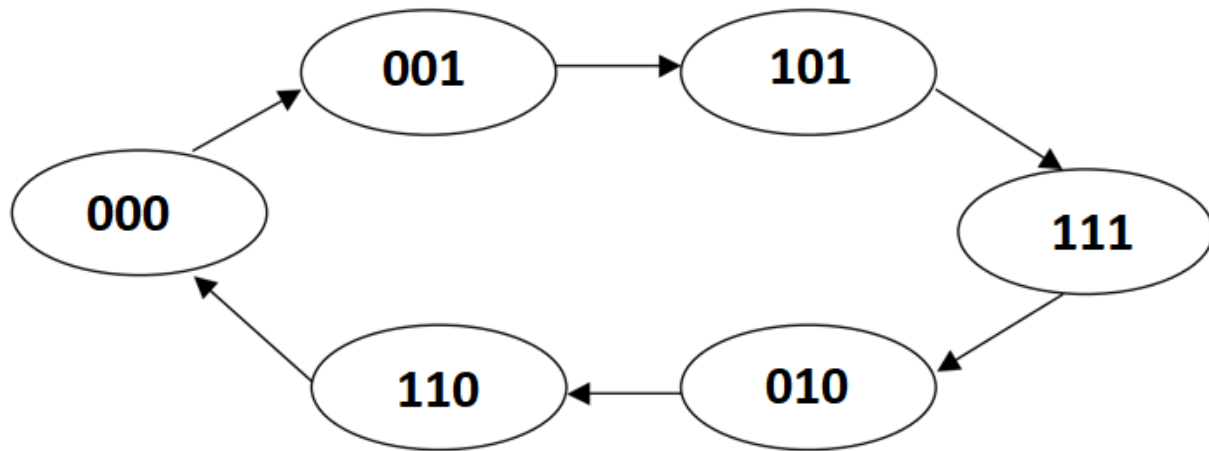
1. Design and build a **minimum counter** circuit with the count sequence given to you.
2. Use **D flip-flops** for your design.
3. Make sure that unused states are designed to go to one of the states in the intended count sequence. Demonstrate this to the TA by forcing the counter (Use **PRE** and **CLR** inputs to do this) to start in each of the unused states.
4. Your report should consist of (1) transition graph, (2) transition table, (3) K-maps, (4) circuit equations, (5) the circuit diagram, and (6) the details about debugging.

GRADING FOR THE LAB EXAM:

Grade over 100

1. Incorrect design on paper10
2. Correct design on paper but not optimal20
3. Correct design on paper, optimal solution30
4. Correct, not optimal design on paper **and** circuit built, but not working correctly35
5. Correct, optimal design on paper **and** circuit built, but not working correctly40
6. Correct, not optimal design on paper **and** circuit built, and working correctly70
7. Correct, optimal design on paper **and** circuit built, circuit working correctly85
8. **NOT optimal design** on paper, circuit working correctly, and successful debugging90
9. **Optimal design** on paper, circuit working correctly, **and** successful debugging100

VERSION A: if the last digit of your PSID is (0, 1, 2, 3, 4)



VERSION B: if the last digit of your PSID is (5, 6, 7, 8, 9)

